Darsh Bhatt

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PROFILE

Software engineer with experience building Python-based data systems, scalable cloud workflows, and RAG-based LLM applications using vector databases and LangGraph. Experienced in building end-to-end solutions spanning data ingestion, retrieval, model interaction, evaluation, and user-facing delivery with Python, SQL, Docker, Kubernetes, and AWS. Strong at translating ambiguous technical problems into practical, reliable solutions for research and operational teams.

CORE COMPETENCIES

Languages: Python, Java, R, SQL, Shell Script
Applied AI / LLMs: LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, LLM Integration, LLM Workflows
ML / NLP: BERT, PyTorch, TensorFlow, spaCy, Scikit-learn
Cloud / Infrastructure: AWS (S3, EC2), Docker, Kubernetes, High-Performance Computing (HPC)
Data / Systems: Pandas, NumPy, Databricks, REST APIs, ETL Pipelines, NoSQL, Oracle Autonomous DB
Frontend / Visualization: React, Power BI
Version Control / Tools: Git, GitLab, RStudio, Jupyter, Azure DevOps

EXPERIENCE

University of Windsor (Research Assistant 2, Windsor, Canada) May 2024 - Present

- Designed containerized analysis workflows using Docker, Kubernetes, and HPC resources to support compute-intensive bioinformatics processing across multi-node environments.
- Built a RAG-based lab assistant using Claude API and vector search over internal documentation to help researchers access technical context through natural-language prompts.
- Improved retrieval and prompt orchestration workflows to reduce manual support by 80% and email-based knowledge transfer, enabling faster self-serve access to internal technical context across the lab.
- Built and optimized Python, R, and SQL pipelines to process and analyze 1M+ phospho-proteomic records, reducing end-to-end runtime by 50% and improving reproducibility across research workflows.
- Developed modular data preparation, transformation, and quality-check pipelines that converted raw biological data into analysis-ready datasets for dashboards, modeling, and reporting.
- Applied machine learning and statistical methods, including clustering, dimensionality reduction, regression, and hypothesis-driven analysis, to high-dimensional biological datasets.
- Used AlphaFold-derived protein structure data and downstream bioinformatics workflows, including Nextflow-based pipelines, to support structural interpretation and residue-level analysis.

ZenQuip (Applied AI Developer, Part-time, Remote, Canada) | *zenquip.com* Jan 2026 - Present

- Built LLM-based workflows for an in-development mental health platform to support conversational interactions and guided user assistance.
- Developed RAG pipelines using LangGraph, vector retrieval, and internal content to improve response grounding and surface relevant support resources.
- Worked with multiple LLM APIs, including OpenAI, Anthropic, and Groq, to test models, refine prompt behavior, and reduce token usage and cost per request by 40% through adaptive RAG workflow tuning.
- Implemented conversation flow logic based on user input to make interactions more context-aware and better aligned with support needs.

TechTrio Solutions (Software Engineering Intern, Ahmedabad, India) Jan 2022 - Jun 2022

- Built Python-based microservices and backend components to support internal applications, data movement, and service-to-service integration.
- Developed Flask REST APIs and integrated them with relational databases to enable secure and scalable data handling for internal tools.
- Containerized backend services with Docker and supported deployment on AWS EC2, improving reliability and reducing setup time across environments.
- Automated SQL- and API-driven data workflows, reducing manual operational effort by 30% and improving consistency across reporting and internal dashboards.
- Collaborated with cross-functional teams in an agile environment to debug production issues, refine requirements, and deliver updates across backend and frontend systems.

EDUCATION

Master of Applied Computing | University of Windsor, Windsor, ON May 2023 - Aug 2024
B.Tech in Computer Science and Engineering | Gujarat Technological University, INDIA Aug 2018 - Mar 2022